Project Proposal Form

(MP3995/7/9 & EI3000)

Part 1: Proposal

Student's Name:	Lowlife555
Module Code:	MP3000
Course Name:	Electrical, Electronics, and Robotics Engineering

Proposed Supervisor	Dr Yu Zhou

Working Title (*provisional*):
Real-Time Webcam Gesture Recognition Using a Neural Network for Robotic Arm Control

Aims and Objectives of the Project:
To develop a webcam-based gesture recognition system, using a neural network, that
controls a robotic arm performing pick-and-place tasks.

1. To train a lightweight neural network to recognise a set of hand gestures from webcam images, achieving at least 90% accuracy on unseen data.

2. To deploy the network on a low-power device for real-time recognition, and measure its accuracy, speed and energy use.

3. To use the recognised gestures to command a robotic arm, and measure the delay from
gesture to completed action and the rate of unintended actions under different lighting and backgrounds.

Brief Introduction of the Proposed Project (*Not exceeding 500 words*)
(including general background, rationale and some brief literature review related to the proposed project)

The whole project: a webcam, a neural network, a development board and a robotic arm, working as one unit. The network learns to tell the gestures apart from camera images. The board runs the network and sends the movement to the arm, which copies it with
its own motor.

The hand can be either a bare hand seen by the camera, or a hand-held device shaped like
the arm.

Outline Plan / Programme of Work / Time Arrangement.

Phase 1 (Oct - Nov 2026): Confirm the robotic arm and laboratory access; literature review; define the gesture set and the dataset; design the system.

Phase 2 (Dec 2026 - Jan 2027): Collect and label image data; train and evaluate the neural network. Non-technical report submitted in January.

Phase 3 (Feb - Mar 2027): Deploy the network on the low-power device; connect it to the
robotic arm; measure accuracy, speed, energy use, delay and unintended actions under
different conditions; analyse the results.

Phase 4 (Mar - Apr 2027): Final evaluation; technical report writing and poster preparation. Draft submitted to the supervisor in March.

May 2027: Poster presentation and viva at the Engineering Expo.

Part 2: Ethics Checklist

All activities - undergraduate, postgraduate, research, commercial, knowledge transfer, evaluation, audit or teaching and learning - need ethical consideration.

This checklist will identify whether a project requires an application for ethics approval, and to which committee it should be referred to. No field work, experimentation or work with participants can start until approval is granted. The questions should be completed in consultation with the supervisor of the proposed project.

Principal Investigators, or supervisors/Director of Studies, are responsible for ensuring that all activities fall within the principles set down in the University Code of Conduct for Research and the University Ethical Principles for Teaching, Research, Knowledge Transfer, Consultancy and Related Activities. They are also responsible for exercising appropriate professional judgment in undertaking this review and evaluating the activity according to the criteria laid down in this checklist. If you are uncertain about any sections of this document, or need further information and guidance, please contact the relevant ethics committee.

If, on completion of the checklist:

- ***any*** question is answered '***Yes***', then an application for ethical approval is required:-
    - For **undergraduate** and **postgraduate taught** projects, students should in the first instance discuss the project and ethical issues with their supervisor. Unless the project is considered to be ethically complex or of a sensitive nature (e.g. involves vulnerable populations) submission for ethical approval should be sought through the relevant School Ethics Committee or process.

- ***all*** questions are answered '***No***' and you (the Principal Investigator) are not concerned with the ethical nature of the activity, then it is unnecessary to apply for ethical approval. However, it is still incumbent on you to observe the University's Ethical Principles in the conduct of the activity and to record that:

    - a review has taken place of the ethical aspects of the activity; and that
    - *either* no ethical issues have been identified *or* ethical issues have been identified but that these have been addressed satisfactorily.

Read any associated procedures and guidance or follow any associated checklist link, and delete, 'Yes' or 'No', for each characteristic.

If you respond 'No', then in your judgment you believe that the characteristic is irrelevant to the activity. You may only tick 'No' to the main question (i.e. A, B, etc) where none of the statements in that section apply to your activity.

If you are unsure whether to answer 'Yes' or 'No' to a question, you should answer 'Yes' and provide details for discussion with your supervisor.

The Ethics Checklist (*in a separate file*) MUST be completed at the same time as this proposal form.

Part 3: Risk Assessment

RISK ASSESSMENT FORM	

Risk Assessment For	Assessment Undertaken By	Assessment Reviewed
Service / School: School of Engineering	**Name:**	**Name:**
Location of Activity:	**Date:**	**Date:**
Activity:	**Signed by Dean of School / Director of Service or their nominee:**	
	Date:	

List significant hazards here:	List groups of people who are at risk:	List existing controls, or refer to safety procedures etc.	For risks, which are not adequately controlled, list the action needed.	Remaining level of risk: H,M or L.

Student Declaration

I confirm that the particulars given in this form are correct. I understand that I must prepare my project in English. I declare that all the work done in connection with my project will be my own.

Signature: _____ Date: _____

Name of the Proposed Supervisor:

Supervisor: if allocated, I agree to supervise this project

Print Name: _____

Signature: _____ Date: _____